Econometria Avançada — MPE (Insper)

Exercise list: *L01 — Review of basic ingredients (book exercises)*

Instructions. Every question is a textbook exercise from the *Practice* list of Class 1 in `Map/class-plan.md`, reproduced verbatim from the book PDF (statements from Portuguese-language books stay in Portuguese). Show every step. Computer exercises are solved in Python; the script `Listas/L01-book-exercises-check.py` reproduces and asserts every number printed in the solutions. Book abbreviations follow the class plan: T = Tsay, W = Wooldridge, G = Greenberg, I = ISLP, B = Bolfarine & Sandoval, MS = Morettin & Singer.

Block 1 — Tsay, *Analysis of Financial Time Series* *(T 2.2)*

Question 1. T 2.2 — AR(2) moments and forecasts.

Source: Tsay, R. S. (2010), *Analysis of Financial Time Series*, 3rd ed., Wiley, Chapter 2 Exercises, Exercise 2.2, p. 104 — reproduced verbatim. (Header of the exercise set: "If not specifically specified, use 5% significance level to draw conclusions in the exercises.")

2.2. Suppose that the daily log return of a security follows the model

$$r_t = 0.01 + 0.2r_{t-2} + a_t,$$

where $\{a_t\}$ is a Gaussian white noise series with mean zero and variance 0.02. What are the mean and variance of the return series r_t? Compute the lag-1 and lag-2 autocorrelations of r_t. Assume that $r_{100} = -0.01$, and $r_{99} = 0.02$. Compute the 1- and 2-step-ahead forecasts of the return series at the forecast origin $t = 100$. What are the associated standard deviations of the forecast errors?

Solution:

Block 2 — Wooldridge, *Introductory Econometrics* *(W 11.C8; W 12.1)*

Question 2. W 11.C8 — AR(1) for unemployment (computer exercise).

Source: Wooldridge, J. M. (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Cengage, Chapter 11, Computer Exercise C8, p. 370 — reproduced verbatim.

C8 Use the data in PHILLIPS for this exercise.

(i) Estimate an AR(1) model for the unemployment rate. Use this equation to predict the unemployment rate for 2004. Compare this with the actual unemployment rate for 2004. (You can find this information in a recent *Economic Report of the President*.)

(ii) Add a lag of inflation to the AR(1) model from part (i). Is inf_{t-1} statistically significant?

(iii) Use the equation from part (ii) to predict the unemployment rate for 2004. Is the result better or worse than in the model from part (i)?

(iv) Use the method from Section 6-4 to construct a 95% prediction interval for the 2004 unemployment rate. Is the 2004 unemployment rate in the interval?

Solution:

Question 3. W 12.1 — OLS standard errors under AR(1) errors.
Source: Wooldridge, J. M. (2016), *Introductory Econometrics: A Modern Approach*, 6th ed., Cengage, Chapter 12, Problem 1, p. 396 — reproduced verbatim.

1 When the errors in a regression model have AR(1) serial correlation, why do the OLS standard errors tend to underestimate the sampling variation in the $\hat{\beta}_j$? Is it always true that the OLS standard errors are too small?

Solution:

Block 3 — Greenberg, *Introduction to Bayesian Econometrics* *(G 2.2; G 7.3)*

Question 4. G 2.2 — gamma–Poisson conjugacy.
Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, §2.5 Exercises, Exercise 2.2, p. 19 — reproduced verbatim.

2.2 The Poisson distribution has probability mass function

$$p(y_i \mid \theta) = \frac{\theta^{y_i} e^{-\theta}}{y_i!}, \qquad \theta > 0,\ y_i = 0, 1, \ldots,$$

and let $y_1, \ldots, y_n$ be a random sample from this distribution.

(a) Show that the gamma distribution $G(\alpha, \beta)$ is a conjugate prior distribution for the Poisson distribution.

(b) Show that $\bar{y}$ is the MLE for θ.

(c) Write the mean of the posterior distribution as a weighted average of the mean of the prior distribution and the MLE.

(d) What happens to the weight on the prior mean as n becomes large?

Solution:

Question 5. G 7.3 — Gibbs sampler for the coal-mining switch point (computer exercise).
Source: Greenberg, E. (2008), *Introduction to Bayesian Econometrics*, 1st ed., Cambridge University Press, §7.5 Exercises, Exercise 7.3, p. 106 — reproduced verbatim. Equations (7.3) and (7.4), to which the exercise refers, are reproduced verbatim from §7.1.1, pp. 93–94.

Equations referred to (Greenberg 2008, pp. 93–94):

$$p(y_i) = \begin{cases} \dfrac{e^{-\theta_1}\theta_1^{y_i}}{y_i!}, & \text{for } i = 1, \ldots, k, \\ \dfrac{e^{-\theta_2}\theta_2^{y_i}}{y_i!}, & \text{for } i = k+1, \ldots, n, \end{cases} \tag{7.3}$$

"where $y_i = 0, 1, \ldots,$ and the switch point k is unknown. The specification is completed by assigning the conditionally conjugate priors,

$$\theta_1 \sim G(\alpha_{10}, \beta_{10}), \quad \theta_2 \sim G(\alpha_{20}, \beta_{20}), \quad \pi(k = j) = 1/n, \quad j = 1, \ldots, n." \tag{7.4}$$

7.3 Construct a Gibbs algorithm to analyze the Poisson model with unknown switch point. Given the specification in Equations (7.3) and (7.4), show that

$$\pi(\theta_1, \theta_2, k \mid y) \propto \theta_1^{\alpha_{10}-1} e^{-\beta_{10}\theta_1} \theta_2^{\alpha_{20}-1} e^{-\beta_{20}\theta_2} \prod_{i=1}^{k} e^{-\theta_1}\theta_1^{y_i} \prod_{i=k+1}^{n} e^{-\theta_2}\theta_2^{y_i},$$

and verify that

$$\theta_1 \mid y, k \sim G\Big(\alpha_{10} + \sum_{1}^{k} y_i,\ \beta_{10} + k\Big),$$

$$\theta_2 \mid k, y \sim G\Big(\alpha_{20} + \sum_{k+1}^{n} y_i,\ \beta_{20} + n - k\Big),$$

$$\pi(k \mid y, \theta_1, \theta_2) = \frac{e^{k(\theta_2-\theta_1)}(\theta_1/\theta_2)^{\sum_1^k y_i}}{\sum_{k=1}^{n} e^{k(\theta_2-\theta_1)}\,(\theta_1/\theta_2)^{\sum_1^k y_i}}, \qquad k = 1, \ldots, n.$$

Apply your algorithm to the mining disaster data analyzed by Carlin et al. (1992a); the data may be found in the article. Your analysis should generate marginal posterior distributions for θ_1, θ_2, $p(k)$, and the marginal likelihood computed by the method of Chib (1995). For simplicity, set $\alpha_{10} = \alpha_{20} = 0.5$ and $\beta_{10} = \beta_{20} = 1.0$. (Curiously, although these data have been analyzed from the Bayesian viewpoint in several papers, none of them present historical material that might shed light on when changes in parameters took place. For example, changes in government regulation, technology, or the location of mines might be expected to affect the probability of a disaster. A more general version of the model permits multiple switch points if necessary.)

Solution:

Block 4 — James, Witten, Hastie, Tibshirani & Taylor, *ISLP* *(I 5.2; I 5.9)*

Question 6. I 5.2 — probability that an observation is in a bootstrap sample.
Source: James, G., Witten, D., Hastie, T., Tibshirani, R. and Taylor, J. (2023), *An Introduction to Statistical Learning with Applications in Python*, Springer, §5.4 Exercises (Conceptual), Exercise 2, pp. 224–225 — reproduced verbatim (the margin index entry "`np.empty()`" is omitted).

2. We will now derive the probability that a given observation is part of a bootstrap sample. Suppose that we obtain a bootstrap sample from a set of n observations.

(a) What is the probability that the first bootstrap observation is *not* the jth observation from the original sample? Justify your answer.

(b) What is the probability that the second bootstrap observation is *not* the jth observation from the original sample?

(c) Argue that the probability that the jth observation is *not* in the bootstrap sample is $(1-1/n)^n$.

(d) When $n = 5$, what is the probability that the jth observation is in the bootstrap sample?

(e) When $n = 100$, what is the probability that the jth observation is in the bootstrap sample?

(f) When $n = 10{,}000$, what is the probability that the jth observation is in the bootstrap sample?

(g) Create a plot that displays, for each integer value of n from 1 to 100,000, the probability that the jth observation is in the bootstrap sample. Comment on what you observe.

(h) We will now investigate numerically the probability that a bootstrap sample of size $n = 100$ contains the jth observation. Here $j = 4$. We first create an array `store` with values that will subsequently be overwritten using the function `np.empty()`. We then repeatedly create bootstrap samples, and each time we record whether or not the fifth observation is contained in the bootstrap sample.

```
rng = np.random.default_rng(10)
store = np.empty(10000)
for i in range(10000):
    store[i] = np.sum(rng.choice(100, replace=True) == 4)
          > 0
np.mean(store)
```

Comment on the results obtained.

Solution:

Question 7. I 5.9 — bootstrap standard errors for Boston `medv` (computer exercise).
Source: James, G., Witten, D., Hastie, T., Tibshirani, R. and Taylor, J. (2023), *An Introduction to Statistical Learning with Applications in Python*, Springer, §5.4 Exercises (Applied), Exercise 9, pp. 227–228 — reproduced verbatim (the margin index entry "`np.percentile()`" is omitted).

9. We will now consider the `Boston` housing data set, from the `ISLP` library.

(a) Based on this data set, provide an estimate for the population mean of `medv`. Call this estimate $\hat{\mu}$.

(b) Provide an estimate of the standard error of $\hat{\mu}$. Interpret this result.
Hint: We can compute the standard error of the sample mean by dividing the sample standard deviation by the square root of the number of observations.

(c) Now estimate the standard error of $\hat{\mu}$ using the bootstrap. How does this compare to your answer from (b)?

(d) Based on your bootstrap estimate from (c), provide a 95 % confidence interval for the mean of `medv`. Compare it to the results obtained by using `Boston['medv'].std()` and the two standard error rule (3.9).
Hint: You can approximate a 95 % confidence interval using the formula $[\hat{\mu}-2\text{SE}(\hat{\mu}), \hat{\mu}+2\text{SE}(\hat{\mu})]$.

(e) Based on this data set, provide an estimate, $\hat{\mu}_{med}$, for the median value of `medv` in the population.

(f) We now would like to estimate the standard error of $\hat{\mu}_{med}$. Unfortunately, there is no simple formula for computing the standard error of the median. Instead, estimate the standard error of the median using the bootstrap. Comment on your findings.

(g) Based on this data set, provide an estimate for the tenth percentile of `medv` in Boston census tracts. Call this quantity $\hat{\mu}_{0.1}$. (You can use the `np.percentile()` function.)

(h) Use the bootstrap to estimate the standard error of $\hat{\mu}_{0.1}$. Comment on your findings.

Solution:

Block 5 — Bolfarine & Sandoval, *Introdução à Inferência Estatística* *(B 1.4, 1.9, 1.12 → 2.2, 2.4, 2.10 → 3.6, 3.13, 3.17)*

Question 8. B 1.4 — MSE of S^2 and $\hat{\sigma}^2$.

Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, §1.4 Exercícios, Exercício 1.4, p. 13 — reproduced verbatim. The expressions it refers to are reproduced verbatim from Exemplo 1.3.6, p. 9.

1.4. Verifique a validade das expressões (1.3.10) e (1.3.11).

Expressões referidas (Exemplo 1.3.6, p. 9): "Sejam $X_1, \ldots, X_n$ uma amostra aleatória da variável aleatória $X \sim N(\mu, \sigma^2)$. Conforme visto no Exemplo 1.3.3, $\hat{\sigma}^2$ é um estimador viciado para σ^2. De (1.3.3) segue que

$$S^2 = \frac{n}{n-1}\hat{\sigma}^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \overline{X})^2$$

é um estimador não viciado para σ^2. Por outro lado, temos (ver Exercício 1.4) que

$$EQM[S^2] = Var[S^2] = \frac{2\sigma^4}{n-1}, \tag{1.3.10}$$

e que

$$EQM[\hat{\sigma}^2] = \frac{2\sigma^4}{(n-1)}\left[1 - \frac{(3n-1)}{2n^2}\right]." \tag{1.3.11}$$

Solution:

Question 9. B 1.9 — Bernoulli: unbiased X versus the constant $1/2$.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, §1.4 Exercícios, Exercício 1.9, p. 14 — reproduced verbatim.

1.9. Seja X uma única variável aleatória com distribuição de Bernoulli com parâmetro θ. Sejam $\hat{\theta}_1 = X$ e $\hat{\theta}_2 = 1/2$ dois estimadores de θ.
(i) Verifique se $\hat{\theta}_1$ e $\hat{\theta}_2$ são não viciados para θ.
(ii) Compare os $EQMs$. Faça um gráfico dos $EQMs$ como função de θ.
Solution:

Question 10. B 1.12 — $U(0, \theta)$: $c_1\overline{X}$ versus $c_2 X_{(n)}$.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, §1.4 Exercícios, Exercício 1.12, p. 14 — reproduced verbatim.

1.12. Sejam $X_1, \ldots, X_n$ uma amostra aleatória de tamanho n da distribuição de uma variável aleatória $X \sim U(0, \theta)$. Considere os estimadores $\hat{\theta}_1 = c_1\overline{X}$ e $\hat{\theta}_2 = c_2 X_{(n)}$.
(i) Encontre c_1 e c_2 que tornam os estimadores não viciados.
(ii) Encontre e compare os $EQMs$ dos dois estimadores.
Solution:

Question 11. B 2.2 — Binomial$(2, \theta)$: Cramér–Rao bound and an efficient estimator.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, §2.6 Exercícios, Exercício 2.2, p. 32 — reproduced verbatim.

2.2. Sejam $X_1, \ldots, X_n$ uma amostra aleatória da variável aleatória $X \sim \mathit{Binomial}(2, \theta)$.
(i) Encontre o limite inferior da variância dos estimadores não viciados de θ.
(ii) Encontre uma estatística suficiente para θ.
(iii) Obtenha um estimador não viciado para θ que seja função da estatística suficiente.
(iv) Verifique se o estimador é eficiente.

Solution:

Question 12. B 2.4 — $X_1 + 2X_2$ is not sufficient for a Poisson mean.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, §2.6 Exercícios, Exercício 2.4, p. 32 — reproduced verbatim.

2.4. Sejam X_1, X_2 uma amostra aleatória da variável aleatória $X \sim \mathit{Poisson}(\theta)$. Mostre que $T = X_1 + 2X_2$ não é suficiente para θ.

Solution:

Question 13. B 2.10 — ENVVUM of $\theta(1-\theta)$ for Bernoulli data.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, §2.6 Exercícios, Exercício 2.10, p. 33 — reproduced verbatim.

2.10. Sejam $X_1, \ldots, X_n$ uma amostra aleatória da variável aleatória. $X \sim \mathit{Bernoulli}(\theta)$. Obtenha o ENVVUM para $\theta(1-\theta)$.
Sugestão: verifique se $S^2 = \frac{n}{(n-1)}\overline{X}(1-\overline{X})$ é não viciado para $\theta(1-\theta)$.

Solution:

Question 14. B 3.6 — MLE of μ^2 versus the unbiased $\overline{X}^2 - 1/n$.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, §3.8 Exercícios, Exercício 3.6, p. 53 — reproduced verbatim. Exercise 2.9, to which it refers, is reproduced verbatim from §2.6, p. 33.

3.6. Encontre o estimador de máxima verossimilhança de θ^2 no Exercício 2.9 e compare seu erro quadrático médio com o do estimador eficiente $\hat{\gamma}$ dado no Exercício 2.9, (i).

Exercício referido (p. 33): **2.9.** Sejam $X_1, \ldots, X_n$ uma amostra aleatória da variável aleatória $X \sim N(\mu, 1)$.
(i) Mostre que o estimador $\hat{\gamma} = \overline{X}^2 - 1/n$ é não viciado para $g(\mu) = \mu^2$.
(ii) Existe ENVVUM para μ^2?
(iii) Encontre o limite inferior da variância dos estimadores não viciados de $g(\mu) = \mu^2$ e verifique se $\hat{\gamma}$ é eficiente.
Note: the parameter of Exercise 2.9 is μ; the "θ^2" of 3.6 is read as μ^2.

Solution:

Question 15. B 3.13 — MLE of $P[X > 1]$ for exponential data and its large-sample distribution.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, §3.8 Exercícios, Exercício 3.13, p. 54 — reproduced verbatim.

3.13. Sejam $X_1, \ldots, X_n$ uma amostra aleatória da variável aleatória X com distribuição exponencial com parâmetro θ. Encontre o estimador de máxima verossimilhança de $g(\theta) = P[X > 1]$ e sua distribuição aproximada quando n for grande.

Solution:

Question 16. B 3.17 — consistency of the estimators in five examples.
Source: Bolfarine, H. and Sandoval, M. C. (2010), *Introdução à Inferência Estatística*, §3.8 Exercícios, Exercício 3.17, p. 55 — reproduced verbatim. The list of the examples' estimators below the statement is a summary (not verbatim) of pp. 37, 43, 44 and 51.

3.17. Verifique se os estimadores obtidos nos Exemplos 3.1.2, 3.1.3, 3.2.1, 3.2.3 e 3.6.2 são consistentes.

Estimators in the examples (summary): 3.1.2, Bernoulli(θ): $\hat{\theta} = \overline{X}$; 3.1.3, $U(0, \theta)$: $\hat{\theta} = X_{(n)}$; 3.2.1, Bernoulli(θ): $g(\hat{\theta}) = \overline{X}(1 - \overline{X})$ for $\theta(1 - \theta)$; 3.2.3, $Exp(\theta)$ with density $\theta e^{-\theta x}$: $g(\hat{\theta}) = e^{-1/\overline{X}}$ for $P_\theta[X > 1] = e^{-\theta}$; 3.6.2, gamma$(\alpha, \beta)$ with density $\beta^\alpha x^{\alpha-1} e^{-\beta x}/\Gamma(\alpha)$, method of moments: $\hat{\alpha} = \overline{X}^2/\hat{\sigma}^2$, $\hat{\beta} = \overline{X}/\hat{\sigma}^2$, $\hat{\sigma}^2 = \sum_i (X_i - \overline{X})^2/n$.

Solution:

Block 6 — Morettin & Singer, *Estatística e Ciência de Dados* *(MS 6.15)*

Question 17. MS 6.15 — decomposition of the total sum of squares.
Source: Morettin, P. A. and Singer, J. M. (2021), *Estatística e Ciência de Dados* (versão preliminar, abril 2021), §6.7 Exercícios, Exercício 15, p. 240 — reproduced verbatim. The definitions it uses are reproduced verbatim from §6.2, p. 185.

15) Mostre que $SQTot = SQRes + SQReg$.

Definições (§6.2, p. 185): "[...] em que a **soma de quadrados total** é $SQTot = \sum_{i=1}^{n}(y_i - \overline{y})^2$, a **soma de quadrados dos resíduos** é $SQRes = \sum_{i=1}^{n}(y_i - \widehat{y}_i)^2$ e a **soma de quadrados da regressão** é $SQReg = \sum_{i=1}^{n}(\widehat{y}_i - \overline{y})^2$." (Model: $y_i = \alpha + \beta x_i + e_i$, with $\widehat{y}_i = \widehat{\alpha} + \widehat{\beta} x_i$ fitted by least squares.)

Solution: